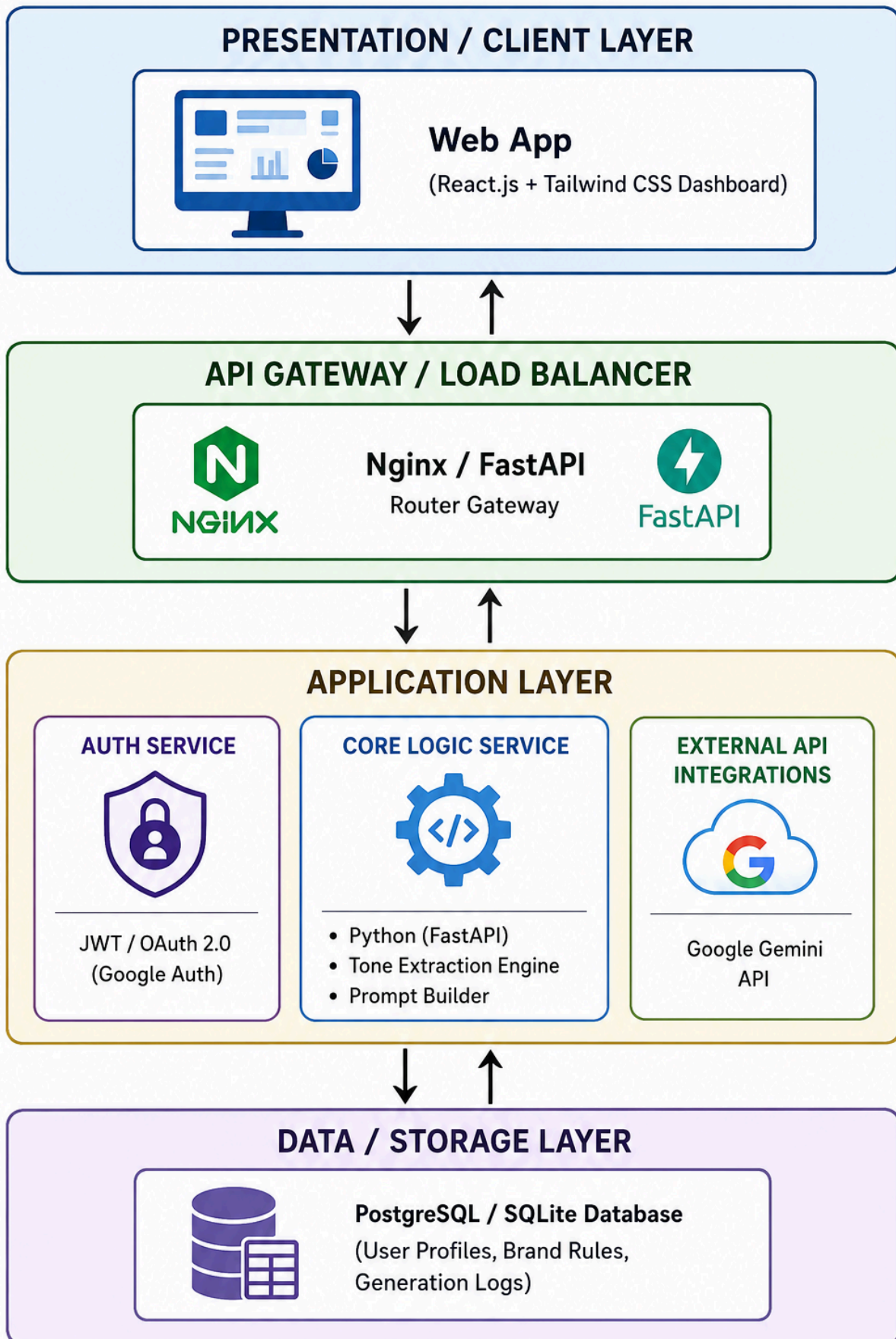


Solution Architecture

Date	12 August 2026
Name	Chahat Chouhan
Team ID	Individual Project
Project Name	AI-Brand-Voice-Generator
Maximum Marks	5 Marks

Solution Architecture Diagram



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides an interactive, user-friendly dashboard for uploading brand samples, configuring tone profiles, entering content prompts, and rendering AI-generated copy in real-time	React.js, Tailwind CSS, HTML5/JavaScript
API Gateway	Acts as the single entry point for incoming client requests, handling request routing, rate limiting, and securing backend endpoints.	Nginx, FastAPI Route Handler
Microservices / Core Services	Manages user authentication/authorization, extracts key brand tone attributes from raw text, constructs context-rich prompts, and interfaces with external AI models.	Python (FastAPI), JWT, OAuth2, Google Gemini API SDK
Database	Stores persistence data including user credentials, defined brand voice rules/guidelines, prompt templates, and historical content generation records	PostgreSQL / SQLite